Jie Cao



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INTRODUCATION

My research interests are mainly in millimetre-wave radar systems and artificial intelligence (AI) algorithms in microwave sensing, especially in biomedicine, human-computer interaction (HCI) and indoor localization. For microwave sensing, I am concerned with the construction of machine learning models with good interpretability and high robustness, including the custom pattern recognition models dominated by kernel method and Bayesian inference, and the construction of multimodal big models driven by neural network.

EDUCATION

Shanghai Jiao Tong University (SJTU) **5 years** **Sep. 2023 – Present**
Ph.D. in information and communication engineering, advised by Prof. Changzhan Gu

Harbin Institute of Technology, Shenzhen (HITSZ) **4 years** **Aug. 2019 – Jun. 2023**
B.E. in communication engineering

RESEARCH EXPERIENCE

PhD student **Shanghai Jiao Tong University (Shanghai)**
Millimetre-wave radar systems, AI algorithms in microwave sensing Sep. 2023 – Present

Intern **Shanghai Artificial Intelligence Laboratory (Shanghai)**
Millimeter-wave radar-based non-contact gesture interaction Aug. 2023 – Jul. 2024

Intern **Open Innovation Lab, DAIKIN (China) Investment Co., Ltd. (Shenzhen)**
Visual recognition algorithms, Biosensor algorithms Jun. 2023 – Jul. 2023

GRANTS &AWARDS

• Mar. 2026, Ceyear Scholarship SJTU, Shanghai, China
• Jul. 2025, 2rd Prize of Best Student Paper Award IEEE MTT-S IMWS-AMP, Wuxi, China
• May 2025, 3rd Prize of Best Student Paper Award IEEE MTT-S IWS, Xi'an, China
• Sep. 2022, National Scholarship HITSZ, Shenzhen, China

PROFESSIONAL ACTIVITIES

• Reviewer for IEEETMTT / JBHI / TIM
• 2025 Outstanding reviewer for IEEE TIM

PUBLICATIONS

Journal paper:

[1] **Jie Cao**, Jingtao Liu, Keke Zheng and Changzhan Gu, "Accurate Phase Accumulation Technique under Vector Cross-multiplication Framework for High-Accuracy Displacement Motion Sensing," *IEEE Transactions on Microwave Theory and Techniques*, vol. 73, no. 11, pp. 9279-9291, Nov. 2025.

[2] **Jie Cao**, Li Wen, Shuqin Dong, Yuchen Li, Linsheng Wu, Yijing Guo, Zi Zeng, Qing Cao, Kang Chen and Changzhan Gu, "Contactless Cardiac Detection of Arrhythmia based on Distribution Encoding Network with Microwave Biomedical Radar," *IEEE Transactions on Microwave Theory and Techniques*, vol. 74, no. 4, pp. 3135-3148, Apr. 2026.

[3] **Jie Cao**, Sitong Li, Shuqin Dong, Xiaoling Wang, Yijing Guo, Min Zhu and Changzhan Gu, "Radio Frequency Sensing with Genetic Manifestations for Assessing Circadian Rhythm Dysregulation from Mouth Breathing," *Electromagnetic Science*, Oct. 2026.

[4] 曹杰，温力，唐敏，郭宜竞，顾昌展*, "基于匹配滤波的非接触生命体征测量," 电子测量技术,

2025,48(01):39-45. (北大核心)

[5] Li Wen, **Jie Cao**, Yijing Guo, Zi Zeng, Qing Cao, Kang Chen and Changzhan Gu, "Robust and Adaptive DC Offset Calibration in Microwave Sensing Systems for Long-Term Microwave Cardiogram Detection," *IEEE Transactions on Microwave Theory and Techniques*, vol. 74, no. 2, pp. 1756-1769, Feb. 2026.

[6] Yuchen Li, **Jie Cao**, Qishan Chen, Li Zhang, and Changzhan Gu, "Unobtrusive Cardiac Activity Monitoring Using Biomedical Radar: Principles, Applications, and Perspectives," *Bioelectromagnetics in Medicine*, vol. 1, no. 1, Aug. 2026.

Conference paper:

[1] **Jie Cao**, Li Wen, Liling Wang and Changzhan Gu, "Contactless Detection of Atrial Fibrillation in Stroke Patients Based on Bayes Probability Model with Microwave Biomedical Radar," *IEEE International Microwave Biomedical Conference (IMBioC)*, Cosenza, Italy, Apr. 26-28, 2026.

[2] **Jie Cao**, Li Wen, Shuqin Dong, Min Tang, Yijing Guo, Chengxing Shen and Changzhan Gu, "Arrhythmia Classification Based on Distribution Encoding Network with Biomedical Radar," *IEEE MTT-S International Microwave Workshop Series on Advanced Materials and Processes for RF and THz Applications (IMWS-AMP)*, Wuxi, China, Jul. 23-26, 2025. **(2nd Prize of Best Student Paper Award)**

[3] **Jie Cao**, Shuqin Dong, Sitong Li, Xiaoling Wang, Min Zhu and Changzhan Gu, "Microwave Bio-Radar Monitoring of Gene-controlled Activity Changes of Normal and Mouth-breathing Rats," *IEEE MTT-S International Wireless Symposium (IWS)*, Xi'an, May 19-22, China, 2025.

[4] **Jie Cao**, Jingtao Liu, Keke Zheng and Changzhan Gu, "Displacement Motion Detection Based on Accurate Phase Accumulation Technique with Reduced Sampling Rate," *IEEE MTT-S International Wireless Symposium (IWS)*, Xi'an, China, May 19-22, 2025.（**3rd Prize of Best Student Paper Award**）

[5] **Jie Cao**, Yuchen Li, Jingtao Liu, and Changzhan Gu,"A Novel Contactless Concentration Evaluation Technique Using Millimeter-Wave Radar," *2024 International Conference on Microwave and Millimeter Wave Technology (ICMMT)*, Beijing, China, May 16-19, 2024.

[6] Yuchen Li, Zichen Huang, **Jie Cao**, Xiaolei Xu, Qishan Chen, Li Zhang and Changzhan Gu, "Millimeter-Wave Radar-Based Unobtrusive Arrhythmia Classification Using Deep Learning," *2026 56th European Microwave Conference (EuMC)*, London, UK, Oct. 6-8, 2026.

[7] Yuchen Li, Zichen Huang, **Jie Cao**, Xiaolei Xu, Qishan Chen, Li Zhang and Changzhan Gu, "Non-Contact Cardiac Activity Monitoring via mmWave Radar for Arrhythmia Recognition and STI Estimation," *IEEE MTT-S International Wireless Symposium (IWS)*, Shenzhen, China, May 9-12, 2026.（**2nd Prize of Flash Best Paper Award**）

[8] Yuchen Li, Gaomin Xue, **Jie Cao**, Xiaolei Xu, Qishan Chen, Li Zhang and Changzhan Gu, "A Knowledge-Fusion Deep Learning Approach for Premature Beat Classification Using mmWave Radar in Clinical Settings," *IEEE Asia-Pacific Microwave Conference (APMC)*, Jeju Island, Korea, Dec. 2-5, 2025.

[9] Mingyue Yuan, **Jie Cao**, Liling Wang and Changzhan Gu, "Non-contact Detection of Cheyne-Stokes Respiration and Sighs with Biomedical Microwave Radar," *IEEE MTT-S International Wireless Symposium (IWS)*, Shenzhen, China, May 9-12, 2026. **(IEEE MVC Competition Finalist)**

[10] Li Wen, **Jie Cao**, Zi Zeng, Qing Cao, Kang Chen and Changzhan Gu, "Utilizing Biomedical Radar in Clinical Doppler Cardiographic Detection of Atrial Arrhythmias," *IEEE MTT-S International Wireless Symposium (IWS)*, Xi'an, May 19-22, China, 2025.

[11] Wenzhuo Gao, Lu Meng, **Jie Cao**, Li Wen, Zeng Zi, Qing Cao, Kang Chen and Changzhan Gu, "Noncontact Atrial Fibrillation Detection Utilizing Convolutional Neural Network Based on Doppler Biomedical Radar," *International Conference on Microwave and Millimeter Wave Technology (ICMMT)*, Xi'an, May 19-22, China, 2025.